Learning to Rank Method

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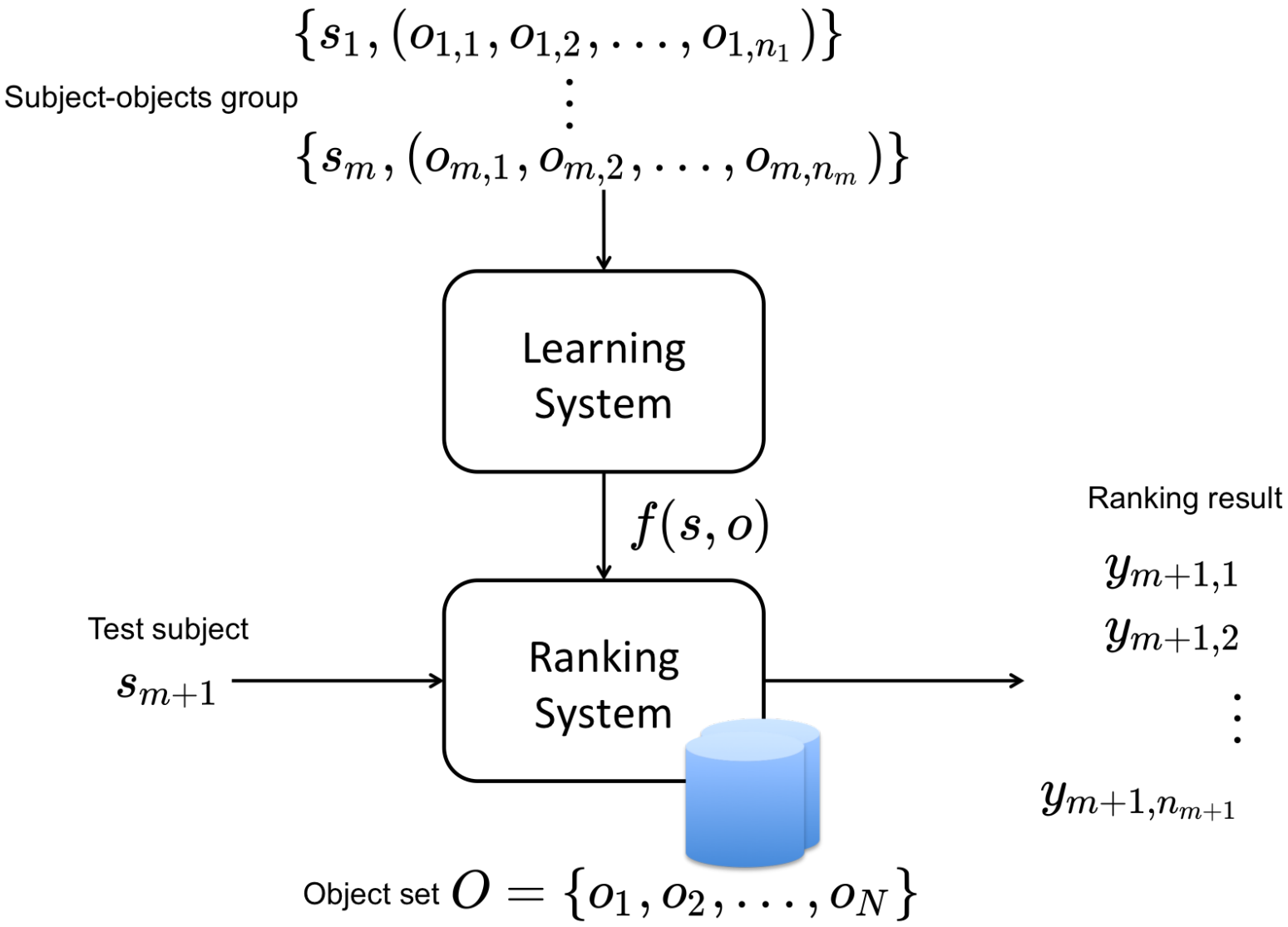
Learning to Rank Method

- A machine learning technique to rank objects
- Ranking depends on both objects and given subject.
- Subject-objects group can be used.
- Applications
  - Search
  - Recommender system
  - Question answering
  - Machine translation

Big Picture: Learning to Rank Method



Learning Process: Learning to Rank Method

$$\text{for } \forall i \in [1, m]$$

$$\begin{array}{ccc} s_i & & s_i \\ o_{i,1} & \xrightarrow{\text{Data Labeling}} & o_{i,1} \;\; y_{i,1} \\ o_{i,2} & & o_{i,2} \;\; y_{i,2} \\ \vdots & & \vdots \\ o_{i,n_i} & & o_{i,n_i} \;\; y_{i,n_i} \end{array}$$

- Pointwise Approach

$$y_{i,j} = f(s_i, o_{i,j})$$

- Pairwise Approach

$$y_m = \begin{cases} 1 & \text{if} \quad o_{i,j} > o_{i,k} \\ -1 & \text{if} \quad o_{i,j} < o_{i,k} \\ 0 & \text{if} \quad o_{i,j} = o_{i,k} \end{cases}$$

- Listwise Approach

$$\begin{bmatrix} y_{i,1} \\ y_{i,2} \\ \vdots \\ y_{i,n_i} \end{bmatrix} = F(s_i, (o_{i,1}, o_{i,2}, \ldots, o_{i,n_i}))$$

Learning Process: Learning to Rank Method

for $\forall i \in [1, m]$

$$
\begin{array}{c}
s_i \\
o_{i,1} \\
o_{i,2} \\
\vdots \\
o_{i,n_i}
\end{array}
\xrightarrow{\text{Data Labeling}}
\begin{array}{cc}
s_i & \\
o_{i,1} & y_{i,1} \\
o_{i,2} & y_{i,2} \\
\vdots & \\
o_{i,n_i} & y_{i,n_i}
\end{array}
\xrightarrow{\text{Feature Extraction}}
\begin{array}{cc}
s_i & \\
x_{i,1} & y_{i,1} \\
x_{i,2} & y_{i,2} \\
\vdots & \\
x_{i,n_i} & y_{i,n_i}
\end{array}
$$

- Relevance feature example

$$x_{i,j} = sim(s_i, o_{i,j})$$
$$x_{i,j} = count(s_i) \text{ in } o_{i,j}$$

- Importance feature example
  - Google's famous PageRank algorithm

Learning Process: Learning to Rank Method

$$\text{for } \forall i \in [1, m]$$

$$\begin{array}{c} \boldsymbol{s_i} \\ o_{i,1} \\ o_{i,2} \\ \vdots \\ o_{i,n_i} \end{array} \xrightarrow{\text{Data Labeling}} \begin{array}{c} \boldsymbol{s_i} \\ o_{i,1}\ y_{i,1} \\ o_{i,2}\ y_{i,2} \\ \vdots \\ o_{i,n_i}\ y_{i,n_i} \end{array} \xrightarrow{\text{Feature Extraction}} \begin{array}{c} \boldsymbol{s_i} \\ x_{i,1}\ y_{i,1} \\ x_{i,2}\ y_{i,2} \\ \vdots \\ x_{i,n_i}\ y_{i,n_i} \end{array} \xrightarrow{\text{Learning}} f(x)$$

Three Approaches of Learning to Rank Method

- Ranking function is a function of features.
- Features can be subject-objects group or objects by subject.
- Three approaches of learning to rank method
  - Pointwise approach

    Transforming ranking to regression, classification, or ordinal regression on single objects
  - Pairwise approach

    Transforming ranking to pairwise classification on object pairs
  - Listwise approach

    Transforming ranking on a object set with given subject

Three Approaches of Learning to Rank Method

	Learning		
	Pointwise Approach	Pairwise Approach	Listwise Approach
Input	Feature vector x	Ordered feature vector pair $(x_i, x_j), x_i > x_j$	Feature vectors $\mathbf{x} = \{x_i\}_{i=1}^n$
Output	$y = f(x)$	$y_{i,j} = classifier(x_i - x_j)$	$\pi = sort(\{f(x_i)\}_{i=1}^n)$
Model		Ranking function $f(x)$	
Loss Function	Regression or classification loss	Pairwise classification loss	Listwise loss

	Ranking
Input	Feature vectors $\mathbf{x} = \{x_i\}_{i=1}^n$
Output	$\pi = sort(\{f(x_i)\}_{i=1}^n)$
Model	Ranking function $f(x)$
Loss Function	Ranking evaluation measure

Pointwise and Pairwise Approaches vs. Listwise Approach

- Pointwise approach ranks on single objects and pairwise ranks on object pairs.
  - Pointwise approach needs a label domain.
  - Pairwise approach needs many compared pairs.
- They can be applied directly.
- But it do not model the ranking problem straightforwardly.
- Listwise approach can model the problem directly.
- Experimental results show the listwise approach usually outperforms other two approaches.

Reference

[1] H. Li. Learning to rank. ACML tutorial, 2010.

[2] F. Xia, T. Liu, J. Wang, W. Zhang, and H. Li. Listwise approach to learning to rank - Theory and Algorithm. ICML, 2008.

[3] R. Nallapati. Discriminative models for information retrieval. SIGIR, 2004.

[4] R. Herbrich, T. Graepel, and K. Obermayer. Support vector vector learning for ordinal regression. ICANN, 1999.

[5] Y. Freund, R. Iyer, R. E. Schapire, and Y. Singer. An efficient boosting algorithm for combining preferences. ICML, 1998.

[6] C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender. Learning to rank using gradient descent. ICML, 2005.

[7] Z. Cao, T. Qin, T. Y. Liu, M. F. Tsai, and H. Li. Learning to rank: From pairwise approach to listwise approach. ICML, 2007.